United International **University**

School of Science and Engineering

Department of Computer Science and Engineering

Mid Term Exam Trimester: - Summer 2025

Course Title: Fundamental Calculus

Course Code: Math 1151 Marks: 30 Time: 1 hour 30 minutes

[Note that the marks are given in brackets at the end of each question or part question. You have to answer all the questions in order]

Q1 **(a) (i)** Copy and complete the table so that the graph of $y = f(x)$ is symmetrical about y axis

x	-3	-2	-1	0	1	2	3
y	7	?	2	1	?	5	?

2 +2

(ii) Verify the following functions as even or odd or neither.

$f(x) = -xcosx$ and $g(x) = 1 + 2e^{-x^2}$

(b) Sketch the graph of following function and find its domain and range. 2+2

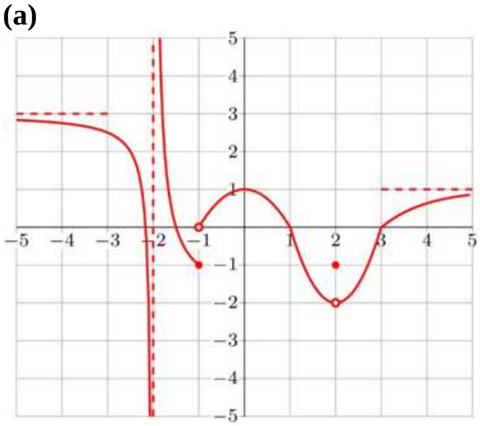
$$y = f(x) = \begin{cases} 2 - x; & -2 \le x \le 0 \\ 2\,; & 0 < x \le 2 \\ -x^2 + 6; & x > 2 \end{cases}$$

(c) By drawing the graph or otherwise, identify the following functions as one to one or many to one function 2

$(\boldsymbol{i})$ $y = 3 - x^2$

$(\boldsymbol{ii})$ $y = 3e^{-2x} + 1$

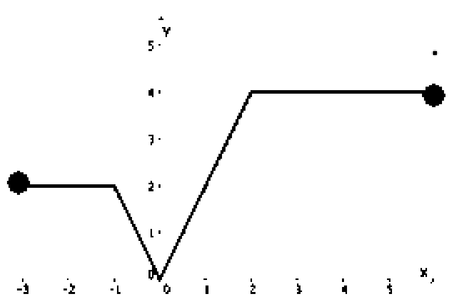
Q2 **(a)**



From the figure answer the following questions:

(i)	The value of $f(2)$	1
(ii)	$\lim_{x \to -2} f(x)$	1
(iii)	$\lim_{x \to -1^-} f(x)$	1
(iv)	$\lim_{x \to 2} f(x)$	1
(v)	Is the function $f(x)$ continuous at $x = 2$? Explain**.**	2

(b) The graph of the curve $y = f(x)$ is drawn below:



(i) Describe in words the transformation represented by $y = 2f(x-1) - 3$. 2+2

(ii) Then, using the above transformation, sketch the graph of the transformed function on the same axes as the original graph $y = f(x)$.

P.T.O.

Q3 **(a)** Let $f(x) = x^2 - x$ and $g(x) = \sqrt{x+3}$. Find $(fog)(x)$ and state the domain of the composite function. 1 +1

(b) Given that $f(x) = \frac{2x}{x-3}, x \neq 3$.

(i) State the equations of the vertical and horizontal asymptotes 2

(ii) Sketch the graph of $f(x)$. 2

(iii) Using your graph, evaluate $\lim_{x \to \infty} f(x)$ 1

(c) Given that $y = 3(x+2)^2 - 1$. The function has inverse for $x \geq a$, state the minimum value of a. Find also the inverse of the function. 1+2